MZUMBE UNIVERSITY
(CHUO KIKUU MZUMBE)

Faculty of Science and Technology
End of Semester II Examinations: July, 2025

Course Code:	**CSS 223**
Course Title:	**Operating System**
Programme:	**BSc. ITS-II, ICTB-II, ICTM-II & EDU (MICT)-II**
Date:	**02/07/2025**
Time:	**0800-1100 HRS**
Venue:	**CHABRUMA-OC & MAREALE-LT 2**

INSTRUCTIONS

1. There are **Four (4)** questions in this paper
2. Answer **All** questions
3. Each question carries **25 Marks**

QUESTION ONE **[25 MARKS]**

a) Explain how the following concepts work in operating system! **(10 Marks)**

   i. Reader-writer –lock

   ii. Compaction

b) Why is spooling important in operating system, and where is applicable? **(8 Marks)**

c) What are the characteristics of Semaphore? **(7 Marks)**

QUESTION TWO **[25 MARKS]**

a) How can buffering improve the performance of a computer system? **(10 Marks)**

b) Explain the concept of pipe in operating system. **(8 Marks)**

c) What is the limitation of pipes for two-way communication between two (2) processes **(7 Marks)**

QUESTION THREE **[25 MARKS]**

a) What is demand paging? **(5 Marks)**

b) How does a thread differ from a process? **(5 Marks)**

c) The following is a step by step program, which involves usage and declaration of semaphore.

```
Shared var mutex: semaphore = 1;
Process i
begin
.
.
```

```
P(mutex);
execute CS;
V(mutex);
.
.
        End;
```

How does this program work? **(15 Marks)**

QUESTION FOUR **[25 MARKS]**

a) Explain the concept of pseudo parallelism. **(5 Marks)**

b) Describe the process model that makes parallelism easier to deal with. **(10 Marks)**

c) Shown below is the workload for 5 jobs arriving at time zero in the order given below. Find out which algorithm among FCFS, SJF and Round Robin with quantum 10, would give the minimum average time, and why. Show all workings clearly in each step. **(10 Marks)**

Job	Burst Time
1	10
2	29
3	3
4	7
4	12